

5	(a) (i)	$I = V/R$	C1	
		$(= 240/1500 =) 0.16 \text{ A}$	A1	[2]
	(ii)	$I_2 = 0.40 - 0.16 (= 0.24)$	C1	
		$0.24(350 + R) = 240$		
		$R = 650 \Omega$	A1	[2]
	(iii)	power = IV or I^2R or V^2/R	C1	
		ratio = $(84 \times 0.24)/(88 \times 0.16)$ or $[(0.24)^2 \times 350]/[(0.16)^2 \times 550]$ or $(84^2/350)/(88^2/550)$ or $20.16/14.08$		
		$= 1.4(3)$	A1	[2]
	(b) (i)	p.d. across 350Ω resistor = 0.24×350 or p.d. across 550Ω resistor = 0.16×550	C1	
		$V_{350} = 84 \text{ (V)}$ and $V_{550} = 88 \text{ (V)}$ gives $V_{AB} = 4.0 \text{ V}$ or $V_{950} = 152 \text{ (V)}$ and $V_R = 156 \text{ V}$ gives $V_{AB} = 4.0 \text{ V}$	A1	[2]
	(ii)	p.d. across R increases or potential at B increases or V_{350} decreases hence V_{AB} increases	B1	[1]